

Mayank Kumar Singh

Full Stack Java Developer — GenAI Enthusiast

[GitHub](#) — [LinkedIn](#) — mayankkumarsingh262004@gmail.com

SKILLS

Languages: Java, Python, C, JavaScript, HTML, CSS

Backend: Spring Boot, REST APIs, Microservices (Basic), Hibernate/JPA

Databases: MySQL, PostgreSQL, MongoDB

Tools: Git, Docker (Basic), Maven, Postman, Linux (Basic)

AI/GenAI: OpenAI API, Gemini API, Ollama (Local LLM), Prompt Engineering

Concepts: OOP, Multithreading, Data Structures & Algorithms, DBMS, OS, Computer Networks, MVC Architecture

COURSEWORK

Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems

PROJECTS

Brain Tumor Detection System with AI Chatbot

[GitHub](#)

- Built an AI-based web application for brain tumor detection using MRI images and machine learning models.
- Integrated Python-based ML model with Spring Boot backend for real-time prediction via REST APIs.
- Implemented an AI-powered chatbot using Ollama (local LLM), enabling offline inference and enhanced user interaction.
- Designed scalable backend services focusing on performance, API efficiency, and real-time data processing.

AI Resume Analyzer using Spring Boot and LLM APIs

[GitHub](#)

- Developed an AI-powered resume analysis platform that evaluates resumes and provides ATS-style feedback using LLM APIs.
- Designed a scalable backend using Spring Boot to process user inputs and generate real-time optimization suggestions.
- Implemented REST APIs for resume parsing, keyword analysis, and structured feedback generation.
- Designed a system to handle concurrent user requests and optimize response time.

AI Chatbot using Spring Boot and Ollama

[GitHub](#)

- Developed an AI chatbot using Spring Boot and Ollama for local LLM-based response generation.
- Implemented REST APIs for handling user queries and delivering real-time conversational responses.
- Optimized response latency and improved user interaction experience.

URL Shortener Service using Spring Boot

[GitHub](#)

- Designed scalable URL shortening service with unique ID generation and redirection system.
- Implemented REST APIs and optimized database queries for high performance.
- Optimized database queries and ensured high availability for fast URL redirection.

EDUCATION

NITRA Technical Campus

Bachelor of Technology (B.Tech)

Final Year (8th Semester)

ACHIEVEMENTS

- Solved 100+ DSA problems on LeetCode focusing on problem-solving, optimization, and coding efficiency.

ADDITIONAL

- Familiar with Linux environment and command-line operations.
- Basic understanding of AWS Cloud (EC2, deployment concepts).
- Strong understanding of system performance, scalability, and reliability.